

Veera Manikanta Gonugondla

Electrical & Electronics Engineer - Hardware Engineer

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SUMMARY

Electrical & Electronics engineering student who designs digital hardware **from the instruction set upward** and gets it running on a board. Current work is MAK8u, a **dual-core microcontroller in SystemVerilog**, closed post-route on an Artix-7 at **50 and 100 MHz** with **126 testbenches** behind it. Also wrote RTL through commercial EDA flows to synthesis on internship, and brought up SPI, I2C and RS-485 with an ILA and a scope. Co-inventor on **two Indian patent applications; Systems & Digital Engineering Intern at Boeing India**.

TECHNICAL SKILLS

Languages	SystemVerilog, Verilog, C, Embedded C, Assembly, Python, MATLAB, Tcl, Bash
Digital Design	RTL design, computer architecture, pipelined in-order cores, hazard detection and forwarding, FSM and datapath design, clock-domain crossing, memory arbitration, interrupt and DMA controllers, ISA definition
Verification	Self-checking SystemVerilog testbenches, automated mutation testing, Coverage, waveform and ILA debug, synthesis and post-route timing closure
Tools	AMD Vivado, Verilator, Nexys A7 (Artix-7), Git, Linux, Simulink, LTspice, LabVIEW, oscilloscopes and logic analyzers, Tcl
Embedded	Raspberry Pi, ESP32, Arduino, FreeRTOS, MQTT, MODBUS RTU / RS-485, PCB-level debugging

EDUCATION

BTech, Electrical & Electronics Engineering

Jul 2023 – Sep 2027

Manipal Institute of Technology, Manipal · **CGPA 8.07 / 10** through 6th semester

Minor Computational Intelligence

BS, Electronic Systems

Jul 2023 – Dec 2027

Indian Institute of Technology Madras · **CGPA 7.33 / 10** through 7th semester

Program Open-entry online BS, taken concurrently with the on-campus BTech at Manipal

CERTIFICATIONS

Computer Architecture Essentials on Arm - Arm Education (Aug 2026)

FPGA Architecture Based System for Industrial Application - L&T EduTech (Aug 2026)

PROJECTS

MAK8u — Asymmetric Dual-Core Microcontroller

Jun 2025 – Present

Overview	A dual-core 8-bit microcontroller built from its own instruction set up. One core runs five stages at 50 MHz for predictable timing, the other seven at 100 MHz for throughput, and one MMCM holds both at an exact 2:1 ratio , so the boundary between them closes under static timing analysis rather than a synchroniser
Feature set	Forwarding and hazard logic on both cores, an inter-core controller with message queues, semaphores and barriers, a lock arbiter that reports the worst-case wait before a core commits to it, DMA, and SPI, I2C, UART, GPIO, PWM and XADC peripherals
Result	Both clocks close post-route, WNS +0.452 ns and +0.480 ns with hold positive, in 8,823 LUTs and 7,619 registers on an Artix-7. Crossing latency held at 2 cycles across 27.5 M silicon measurements , against 32 to 36 asynchronously. Shortened the binding path to the program counter from 14 logic levels to 6
Verification	126 self-checking testbenches , all passing before every synthesis. Five mutation campaigns found the tests that only looked strong
Tooling	A 54-instruction ISA with a Python assembler, ILA capture for silicon debug, and MAK IDE, a terminal workbench that loads images over UART without Vivado
Achievements	Patent draft in progress on repeatable inter-core timing across rationally-related clock domains
Stack	SystemVerilog, Vivado, Tcl, Verilator, Python, Nexys A7 (Artix-7)

PROJECTS (CONT.)

Project Garuda — Edge-AI Home Security & Surveillance

Jan 2025 – Present

Overview	A surveillance stack that runs detection on the device itself. No cloud, and no video leaves the home
Stack	Raspberry Pi 5, Hailo-8L M.2 AI HAT (13 TOPS), YOLOv8s, Python
Result	INT8-quantized inference at 52.2 FPS, 18.4 ms latency and 0.854 mAP@0.5 on a 5.8 W total system power budget
Achievements	Co-inventor, Indian Patent App. No. 202641090505 (filed Jul 2026) · research paper in preparation

Project bREADth — Non-Invasive Multi-Modal Patient Monitor

Mar 2025 – Present

Overview	A wearable monitor for continuous vital signs and early collapse detection, with a macOS app for multi-patient floor-plan tracking
Stack	Multi-modal sensor fusion, DSP, SwiftUI, MQTT, scikit-learn
Result	Embedded Random Forest fall detection over 104 features , plus a local RAG agentic chatbot for vital-sign diagnostics and clinical handbook querying
Achievements	Co-inventor, Indian Patent App. No. 202541059846 (published Aug 2026) · manuscript under review at IEEE Sensors Journal .

8-Channel Thermal Profiling & Heatmap System

Feb 2025

Overview	Eight DS18B20 sensors on a single 1-Wire GPIO, read by an ESP32-S3 exposed as a MODBUS RTU slave over RS-485
Stack	ESP32-S3, DS18B20 / 1-Wire, MODBUS RTU over RS-485, Python, PyQt5, scikit-learn, SwiftUI
Result	A macOS SCADA dashboard with live 2D heatmaps, 3D surface plots, hotspot analysis, CSV logging and real-time Isolation Forest anomaly detection

Active Battery Cell Equalization for EV Applications

Sep 2025 – Oct 2025

Overview	An active BMS using a bidirectional flyback converter to move charge between series-connected Li-ion cells instead of dissipating it resistively
Stack	MATLAB, Simulink, power electronics
Result	A proportional duty-cycle controller with dynamic threshold monitoring drove peak inter-cell divergence from ~1.3 V to near zero , versus persistent imbalance under passive balancing

WORK EXPERIENCE

Systems & Digital Engineering Intern, Boeing India Private Limited

May 2026 – Jul 2026

- Built a fleet maintenance analytics platform (Python, FastAPI, React, TypeScript) that answers queries **across 6,900+ maintenance records in under a second**, with a RAG chatbot on a 5-stage deterministic NLQ pipeline and automated reporting.
- Wrote a forecasting engine that picks among 11 algorithms by horizon over a 24-feature pipeline, cutting error **up to 50%** against linear interpolation.
- Trained an 82-feature XGBoost failure-prediction model (**F1 0.964, ROC-AUC 0.952**) with SHAP explainability; its risk alerts save **40+ engineering hours a month**.

Project Intern, Mindenious Edutech (Remote)

Jul 2025 – Oct 2025

- Completed a VLSI / ASIC design program on commercial EDA tools.
- Wrote RTL in Verilog / SystemVerilog with self-checking test benches, through synthesis to FPGA.
- Measured functional coverage in simulation before taking designs to synthesis.
- Analyzed timing and area trade-offs across synthesis constraints on FPGA targets.

Project Intern, Apsis Solutions & IIT Guwahati (Remote)

Jul 2025 – Oct 2025

- Built a factory-automation monitoring system covering 13 machines. Manual check time fell **89%**.
- Instrumented the floor for continuous machine-status capture in one live supervisor view.
- Coordinated with IIT Guwahati mentors to validate monitoring accuracy against manual floor logs.

Summer School, Product Development Management, IIIT Hyderabad

Jul 2024 – Aug 2024

- Studied product lifecycle, market analysis and innovation strategy, then carried one concept from market study to a pitched prototype.
- Presented the prototype and go-to-market plan to a panel of mentors and industry reviewers.

ACHIEVEMENTS

1st Place, IITM Gadget Expo — Best Low-Cost Solution, for Project Garuda
Co-inventor on **two Indian patent applications** filed for Project Garuda and Project bREADth